Fuzzy logic (A) AND (B)

A \ B	0	0.25	0.5	0.75	1
0	0	0	0	0	0
.25	0	0.25	0.25	0.25	0.25
0.50	0	0.25	0.5	0.5	0.5
0.75	0	0.25	0.5	0.75	0.75
1	0	0.25	0.5	0.75	1

Ans

Fuzzy logic A OR B

A \ B	0	025	0.5	0.75	1.0
0	0	0.25	0.5	0.75	1.0
0.25	0.25	0.25	0.5	0.75	1.0
0.5	0.5	0.5	0.5	0.75	1.0
0.75	.75	0.75	0.75	0.75	1.0
1.0	1.0	1.0	1.0	1.0	1.0

Ans

2 (c) Show the following fuzzy set operations

i) A∪B (ii) A∩B (iii) B\A (iv) A^c

(v) A is subset of B

$A = (0.4, 0.6, 0.7)$; $B = (1.0, 0.3, 0.4)$

Ans: (i) $A \cup B = (0.4, 0.6, 0.7) \cup (1.0, 0.3, 0.4)$

$= (1.0, 0.6, 0.7)$ Ans:

(II) $A \cap B = (0.4, 0.6, 0.7) \cap (1.0, 0.3, 0$

$= (0.4, 0.3, 0.4)$ Ans:

(III) $B \setminus A = (1.0, 0.3, 0.4) \setminus (0.4, 0.6, 0.7)$

$= (0.6, 0, 0)$ Ans:

(IV) $A^c = (0.4, 0.6, 0.7)^c$

$\therefore A^c = (0.6, 0.4, 0.3)$ Ans:

(V) A is subset of B

$= A \subset B$

$= (0.4, 0.6, 0.7) \subset (1.0, 0.3, 0.4)$

$= (YES, NO, NO)$ Ans: